

Note from Flint:

Some research projects are novel not because of a new algorithm or tighter theorem, but because they identify, formalize, and defend an important problem that has not been studied rigorously before. This is a legitimate — often harder — form of contribution. Many foundational research directions begin exactly this way. FedHQL and FedRLHF are examples: their novelty rests partly on elevating an underdefined practical concern into a precise, analyzable research object.

Use this supplement alongside the Playbook, not instead of it. The Playbook structure applies. What changes is the emphasis, the evaluation standard, and what belongs where.

A. Why This Contribution Is Legitimate — and Hard

A1. Formulation can be the central novelty

The contribution is transforming a vague practical concern into a rigorous, reproducible scientific object. Do not worry if your formulation brings more questions than it answers — that is a sign of success, not incompleteness. A good formulation generates a research agenda. A closed exercise does not.

A2. The core work is abstraction, not ideation

The hard part is not having the idea — it is building the right abstraction. You must decide: what is the essential phenomenon, which aspects of reality must be preserved, which simplifications are acceptable, and how to make the setup both meaningful and analyzable. A strong formulation is disciplined, not arbitrary.

A3. Realism vs. tractability is a design trade-off

Too abstract: looks disconnected from practice. Too realistic: too messy to analyze. The goal is the middle ground: realistic enough to matter, structured enough for rigorous analysis and empirical validation. You choose where to draw this line — and you must defend that choice explicitly.

A4. Being first gives freedom, but also burden

The standard is not “anything goes because nobody has done it before.” The standard is: if you define the problem first, you must define it convincingly. This is usually where formulation papers face the hardest scrutiny.

B. Core Requirements Before You Commit

B1. Verify the gap is real

Confirm no existing paper defines this problem in a clean and rigorous way. This requires a systematic survey (as many papers as possible, but from venues we recognize) across all plausible neighboring areas. You need explicit gap evidence: papers that either identify the problem as open, use a clearly inadequate proxy, or operate under assumptions that exclude your setting.

B2. Confirm tractability

A formulation you cannot study is not a contribution. Before committing, check:

- Can you write a clean optimization objective or protocol?
- Can you instantiate it in an existing environment or dataset?
- Can you derive at least one non-trivial theoretical property, even under simplifications?

B3. The deliverable is a formal definition

Compelling motivation is necessary but not sufficient. At some point the paper must contain a precise formal object: a tuple, a loss function, a set of constraints, a protocol. If you cannot write that object down, you have not formulated the problem — you have only described it. Everything else (motivation, related work, experiments) is there to justify and validate the formal definition.

Self-test: give your Problem Formulation section to a fellow student not on the project. Ask them to reconstruct the setup from scratch. If they cannot, the section is (likely) incomplete.

C. Adapting the Playbook Template

The standard Playbook structure applies. Here is what changes section by section:

Section	Standard paper	Formulation paper: what changes
Abstract S3 (Key idea)	<i>The mechanism of your algorithm</i>	The formal structure of your problem definition — name its essential components, not an algorithmic step.
Abstract S4 (Results)	<i>Theory + empirical gains with numbers</i>	Theory = a structural property of your formulation (lower bound, impossibility, tractability). Empirical = evidence the problem instantiates and is non-trivial.
Intro: Contributions	<i>3–5 algorithmic/theoretical contributions</i>	Lead with: (1) the formal definition, (2) the gap it closes, (3) structural properties or theory. Algorithmic results follow as supporting evidence.
Related Work table	<i>Columns = algorithmic properties</i>	Add a column: "Formally defines the problem class (Y/N)." Most prior work will be N. This is your primary differentiator.

Section	Standard paper	Formulation paper: what changes
Problem Formulation	<i>Setup section preceding the method</i>	This is now a primary contribution section. Longer, more careful, fully self-contained. Must include: all notation, formal objective, evaluation metric, assumptions with justification, explicit scope boundaries.
Method	<i>Central technical contribution</i>	May be demoted. If present, treat it as a first baseline, not the headline. An analysis of structural properties of the formulation can replace it.
Theory	<i>Convergence/regret of your algorithm</i>	First target: a structural result about the problem class (hardness, lower bound, tractability condition). Algorithm analysis is secondary.
Experiments	<i>Benchmark method against baselines</i>	Primary purpose: validate the formulation. At minimum, include one experiment designed to answer: is this problem real and non-trivial under plausible conditions?

D. Defending the Formulation Against Reviewers

D1. Main attack: “This is just X reframed”

You cannot defend with rhetoric. You must show structurally: identify the specific property of your formulation that is absent from the closest prior works, and demonstrate that this property creates a genuine technical obstacle prior methods cannot address without fundamental modification. Your Related Work table is the primary vehicle for this.

D2. Secondary attack: “The assumptions are too strong”

For each assumption, prepare two sentences: (1) why it is necessary for your analysis, and (2) what evidence supports its reasonableness in the target setting. Write these defenses in the paper — do not save them for the rebuttal.

D3. Experiments must validate the formulation, not just the method

Ask yourself: if someone read only the experiments section, would they conclude the formulated problem is real and important? If not, redesign at least one experiment to directly demonstrate the problem manifests and is non-trivial under plausible conditions.

E. Fill-In Language Snippets

- **Problem formulation statement.** “We study [phenomenon] in [setting]. Unlike prior work, which treats [related problem] under [prior assumption], we formalize a setting in which [distinguishing structural property]. Formally: [definition].”

- **Gap framing (formulation-specific).** “Prior work addresses [related problem], but no existing formulation captures [structural property X] that arises when [condition]. This gap is not merely notational: [specific obstacle] cannot be addressed by [prior method] without [fundamental modification].”
- **Contribution item: the formulation itself.** “We introduce [Problem Name], the first formalization of [phenomenon] under [setting]. Our definition captures [property A] and [property B], which are absent from prior formulations of [related work].”
- **Assumption justification.** “[Assumption X] ensures [tractability property]. While this simplifies [real-world complexity Y], the essential structure of the problem is preserved: [argument]. Relaxing this assumption is left to future work; we scope this in Section [Z].”
- **Experiment framing: formulation validation.** “The objective of this experiment is to verify that [formulated problem] manifests under plausible conditions. We instantiate our formulation in [environment] with [parameters] and show that [phenomenon] is present and non-trivial, confirming the formulation captures a genuine challenge beyond [prior proxy].”

F. Evaluation Criteria for This Genre

Reviewers judge formulation papers differently. The key questions are not about benchmark wins but:

- **Is the problem important?** → Motivation + real deployment scenario in Introduction
- **Is the formulation precise?** → Problem Formulation section: formal definition, full notation
- **Are the assumptions sensible?** → Per-assumption justification in Theory section
- **Is the setup realistic enough to matter?** → Formulation-validation experiment
- **Is this genuinely new, or just X reframed?** → Related Work table with "formally defines problem class" column
- **Does this make the problem easier for the community to study?** → Discussion: open questions generated by the formulation

After publication, a clear formulation often becomes the reference point for an entire line of subsequent work. Later papers build on it, critique it, or replace it — but they must position themselves against it. That is the long-term value of this kind of paper, and why it is worth doing carefully.
